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Antagonist muscle moment is increased in ACL deficient subjects during maximal dynamic knee extension

Tine Alkjær [a,*], Erik B. Simonsen [a], S. Peter Magnusson [b,c], Poul Dyhre-Poulsen [a,1], Per Aagaard [d]

[a] Department of Neuroscience and Pharmacology, University of Copenhagen, DK-2200 Copenhagen N, Denmark
[b] Institute of Sports Medicine, Department of Orthopaedic Surgery, Bispebjerg Hospital, and Faculty of Health Sciences, University of Copenhagen, Denmark
[c] Department of Kinesiology, The University of North Carolina at Charlotte, Charlotte, NC, USA
[d] Institute of Sports Science and Clinical Biomechanics, University of Southern Denmark, Odense, Denmark

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ABSTRACT

Introduction: Coactivation of the hamstring muscles during dynamic knee extension may compensate for increased knee joint laxity in anterior cruciate ligament (ACL) deficient subjects. This study examined if antagonist muscle coactivation during maximal dynamic knee extension was elevated in subjects with anterior cruciate ligament (ACL) deficiency compared to age-matched healthy controls.
Methods: Electromyography (EMG) and net knee joint moments were recorded during maximal concentric quadriceps and eccentric hamstring contractions, performed in an isokinetic dynamometer (ROM: 90–10°, angular speed: 30°/s). Hamstring antagonist EMG recorded during concentric quadriceps contraction was converted into antagonist moment based on the EMG–moment relationship observed during eccentric agonist contractions.
Results: The magnitude of antagonist hamstring EMG was 65.5% higher in ACL deficient subjects compared to healthy controls ($p<0.05$). Likewise, antagonist hamstring moment expressed in percentage of the measured net extension moment was elevated in ACL deficient subjects (56 ± 8 to 30 ± 6%) compared to controls (36 ± 5 to 19 ± 2%) at 20–50° of knee flexion (0° = full extension) ($p<0.05$).
Discussion: The results showed a marked increase in hamstring coactivation towards more extended joint positions. Notably, this progressive rise in coactivation was greater in ACL deficient subjects, which may reflect a compensatory strategy to provide stability to the knee joint in the anterior–posterior plane during isolated knee extension. The present study encourages further investigations of hamstring coactivation in ACL deficient subjects.

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1. Introduction

It has been demonstrated that the quadriceps muscle can cause anterior translation and internal rotation of the tibia during both isometric quadriceps contraction and in dynamic knee extension [1–7]. The primary mechanical function of the anterior cruciate ligament (ACL) is to limit anterior translation of the tibia relative to the femur. Thus, during movements that include knee extension the shear force will be transmitted to the ACL in the intact knee and in case of ACL deficiency this will cause substantial anterior tibial translation (ATT) [7]. Increased ATT in ACL deficient subjects may increase the risk of giving way in dynamic situations which impairs the functional performance and it may also cause injury to other knee joint structures such as the menisci [8]. Thus, in case of ACL insufficiency it seems crucial to minimize ATT in order to stabilize the knee and thereby protect other knee joint structures from overuse injury. Increased coactivation of the hamstring muscles has been suggested to provide a compensatory strategy that ACL deficient subject may use to reduce ATT in functional conditions that include knee extension [4,7,9]. Several studies have shown that the hamstring muscles are active during submaximal and maximal quadriceps agonist contractions [9–14] and that coactivation of the antagonist hamstring muscles during knee extension effectively reduces the amount of ATT [7,13,15–19].

The contractile potential for the hamstring muscles to stabilize the knee joint has traditionally been evaluated based on the ratio between concentric hamstring and quadriceps strength [20]. Alternatively, Aagaard et al. [21] suggested that the potential for muscular stabilization during knee extension was better functionally described by the ratio of maximal eccentric hamstring strength to concentric

* Corresponding author at: Department of Neuroscience and Pharmacology, University of Copenhagen, Blegdamsvej 3, DK-2200 Copenhagen N, Denmark. Tel.: +45 35 32 72 16.
E-mail address: talkjaer@sund.ku.dk (T. Alkjær).
[1] Deceased.

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quadriceps strength; denoted by the so-called functional H/Q ratio representative for isolated knee extension. In contrast to conventional H/Q ratio, data on functional H/Q demonstrated that the potential of the hamstrings to stabilize the knee joint was markedly elevated at more extended knee joint positions [21]. However, only few studies have estimated the magnitude of antagonist hamstring muscle moment during actual knee joint extension. In these previous studies the antagonist hamstring moment was estimated by a model based on the EMG–moment relationship in which the antagonist EMG recorded during active knee extension (i.e. *eccentric* contraction of the hamstrings) was normalized to the EMG obtained during maximal *concentric* agonist contraction of the hamstrings [12,22]. Thus, antagonist moments during eccentric muscle actions were estimated from concentric EMG–moment relationships, which likely underestimated the antagonist hamstring moments. Alternatively, estimates of antagonist hamstring moments based on actual eccentric EMG–moment calibration have been reported as well [10,23].

A number of studies have reported increased coactivation of the hamstring muscles in ACL deficient subjects during movements like walking, and have interpreted this as a strategy to stabilize the knee and compensate for the increased joint laxity [14,24–26]. However, only few studies have investigated the magnitude of hamstring coactivation and the resulting production of antagonist moment during isolated isokinetic knee extension in ACL deficient subjects. The results from these studies suggest that the level of hamstring coactivation in ACL deficient patients is comparable to healthy subjects [22,27]. One methodological reason for this may be due to the specific hamstring EMG normalization procedures employed in these studies [10]. Thus, the level of antagonist hamstring EMG was normalized to the maximal EMG amplitude measured during concentric hamstring contraction [22] or expressed relative to the average hamstring EMG activity in the entire ROM [27], which may have systematically affected the obtained data. Therefore, it was the purpose of the present study to examine the magnitude of antagonist hamstring muscle coactivation during maximal voluntary knee extension using an angle-specific and contraction-specific EMG-to-moment model in a group of patients with unilateral ACL deficiency compared to a group of control subjects without knee pathologies. The level of antagonist muscle coactivation was investigated both in terms of EMG recordings and by estimations of the antagonist hamstring moment during isokinetic knee extension in which the individual EMG–moment calibration relationships were established based on the EMG activity and resulting knee flexor moment recorded during eccentric agonist contractions performed by the hamstrings muscles. We hypothesized that hamstring coactivation would increase towards more extended knee joint positions, and that the magnitude of coactivation would be elevated in ACL deficient subjects.

2. Materials and methods

2.1. Subjects

The patient group consisted of seven male subjects with complete unilateral rupture of the anterior cruciate ligament (body mass 76.1 ± 8.3 kg, height 1.80 ± 0.05 m and age 29.6 ± 5.9 years, mean ± SD). The average (± SD) time lapse between injury and testing was 47.5 ± 54.2 months. The circumference of the thigh was measured 15 cm proximal to the patella on both legs of the ACL patients. According to the Tegner score [28] four patients scored 4, one 6 and two 7 indicating a moderate to high activity level. In addition, no statistically significant difference between the circumference of the injured (52.6 ± 2.3 cm) and uninjured thigh (55.1 ± 3.9 cm) was found. The average Lysholm score [29] of the patient group was 78.5 ± 12.8 (range 62–95), indicating that the self-rated knee function within this group ranged from “poor” (one patient) over “fair” (three patients) to “excellent” (three patients).

Sixteen males without any history of previous knee pathologies were selected for the control group in the study (body mass 73.2 ± 5.8 kg, height 1.79 ± 0.04 m, age 23.5 ± 3.4 years, mean ± SD). The control subjects were habitually active with a minimum of 2 h of weekly recreational physical activities and on average the Tegner score was 5 (range 4–6). All subjects gave their informed consent to participate in the experiments, which were approved by the local ethics committee.

2.2. Isokinetic dynamometry

Two types of maximal isolated knee extension movements were performed in an isokinetic dynamometer (Kinetic Communicator, Chattecx Corp., Chattanooga, USA): (i) concentric quadriceps contractions and (ii) eccentric hamstring contractions of maximal voluntary effort. The angular velocity was $30° s^{-1}$ and the range of motion was 90° to 10° (0° = full extension).

The subjects were seated in a rigid chair with a 10° reclined backrest and firmly strapped to the seat at the hip and distal thigh. The rotation axis of the dynamometer was visually aligned to the lateral femoral epicondyle and the lower leg was attached to the lever arm of the dynamometer without static fixation of the ankle joint, which allowed the tibia to rotate during the knee extension. The lever arm was placed 6 cm above the lateral malleolus. Prior to testing, 15 min of warm-up was performed to familiarize the subjects to the dynamometer and the test procedures.

During testing the subjects were asked to perform two types of isokinetic knee extensions: 1) maximal concentric agonist contractions in the quadriceps muscle (hamstring muscles contracting as antagonists) and 2) maximal eccentric agonist contractions of the hamstring muscles (quadriceps antagonist) (Fig. 1). On-line visual feedback of the exerted force was provided to each subject on a PC screen [10,20,21] and successive trials were performed in each contraction mode until the subjects were unable to increase peak moment any further. Typically 6–8 trials were performed to fulfill this criterion. The final three trials performed at each contraction mode (including the trial with the highest recorded moment) were averaged and used for analysis. The KinCom force signal was multiplied by the individual lever arm length to calculate the moment of force (Nm). Subsequently, all recorded moment signals were corrected for the effect of gravity [20].

Both the anterior cruciate ligament deficient (ACLD) and the uninjured (UNI) leg were tested in the ACL deficient subjects. Only the right leg was tested in the control group (CON).

2.3. Electromyography (EMG recordings)

EMG signals were recorded using bipolar surface electrodes (Medicotest Q-10-A and N-00-S, 2 cm inter-electrode distance) placed on vastus lateralis (VL), vastus medialis (VM), rectus femoris (RF), biceps femoris (BFcl) and semitendinosus (ST) [10,30]. The skin was carefully shaved and swabbed with pure alcohol prior to placement of the electrodes, which were connected by shielded wires to custom-built amplifiers with a frequency response of 20 Hz to 10 kHz and a common mode rejection ratio > 100 dB. Small preamplifiers were attached to the electrodes to lower the impedance and eliminate movement artifacts. It was visually confirmed that no cross-talk or movement artifacts were present on the EMG signals before testing. Furthermore, based on cross-correlation analysis, the amount of cross-talk between the quadriceps and hamstring muscles has been shown to be negligible with the present measuring set-up [10].

The subsequent treatment of the EMG signals included digital bandpass filtering (5–200 Hz, Butterworth fourth order zero-lag filters) and smoothing by a symmetric moving-average RMS routine with a time constant of 20 ms [30].

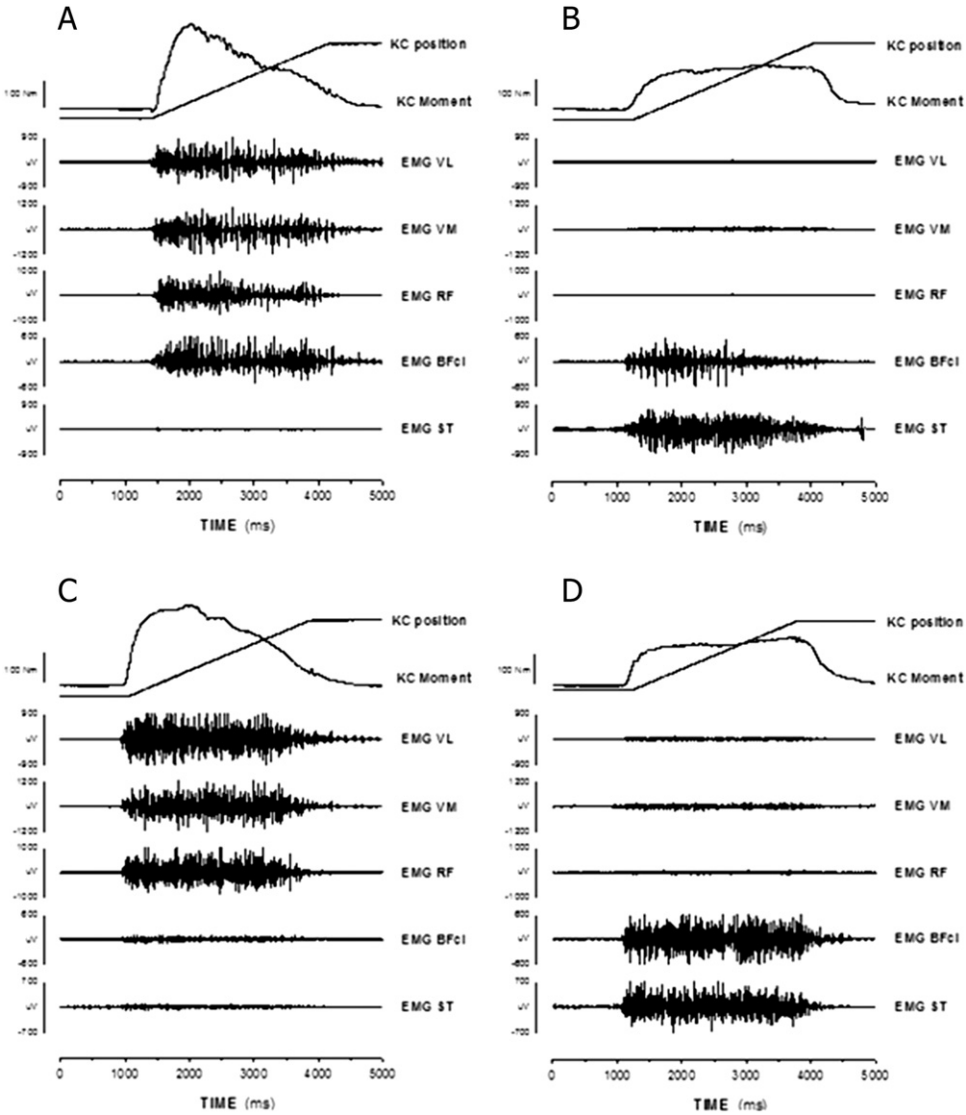

Fig. 1. Representative knee extension trials obtained during agonist contraction of the quadriceps femoris muscle (A,C) and hamstring muscles (B,D), respectively. Top panels (A, B): ACL deficient subject, bottom panels (C, D): uninjured control subject.Top graphs show the net knee extension moment (A, C: KC Moment) and net knee flexion moment (B, D: KC Moment) and knee joint angle (KC position, 90° → 10°, 0° = full knee extension, axis not shown). Bottom graphs show raw EMG signals recorded in the quadriceps (vastus lateralis VL, vastus medialis VM, rectus femoris RF) and hamstring muscles (biceps femoris cap long BFcl, semitendinosus ST). Note that identical contraction modes are employed for each respective muscle across all conditions (A–D), i.e. concentric contraction for the quadriceps muscle (VL, VM, RF) and eccentric contraction for the hamstring muscles (BFcl, ST). Also note that in A, during agonist quadriceps contraction performed by the ACL deficient subject, the EMG of the lateral antagonist hamstring muscle (BFcl) corresponded to about 100% of its agonist EMG (B) with almost no antagonist activity in the medial hamstrings (ST) (<1%). In contrast, the control subject demonstrated low levels of antagonist EMG activity, both for the medial and lateral hamstrings (<15%, C vs D).

2.4. Data acquisition

Synchronisation between the KinCom signals and the EMG signals was obtained electronically. When the subject started to move, the investigator triggered an analogue to digital converter, (dt2801-A converter, Data Translation, Marlboro, MA) which initiated synchronous sampling of the signals at 1000 Hz, using a pre-trigger time of 1.5 s.

2.5. Calculation of antagonist hamstring moment

The eccentric antagonist hamstring muscle moment generated during the agonist quadriceps contraction was calculated according to previously described methods [10]. Briefly, in the trials of maximal eccentric agonist hamstring contraction a relation between the recorded hamstring EMG and the measured net knee flexor moment was established for any given joint angle throughout the ROM. Likewise, during the trials of maximal agonist quadriceps contraction a relation between quadriceps EMG and the measured net moment was established. It was assumed that at any knee joint angle the measured net joint moment was the algebraic sum of the agonist muscle moment minus the antagonist muscle moment. Accordingly, at any given knee joint angle throughout the ROM the measured net knee moment could be described by two equations:

$$M_1 = K_1 \cdot EMG_{Q,agonist} - K_2 \cdot EMG_{H,antagonist[trials\ of\ quadriceps\ agonist,hamstrings\ antagonist]}$$

$$M_2 = K_2 \cdot EMG_{H,agonist} - K_1 \cdot EMG_{Q,antagonist[trials\ of\ hamstrings\ agonist,quadriceps\ antagonist]}$$

Where M_1 was the measured net moment generated during concentric quadriceps contraction (quadriceps contracting as agonist, hamstring contracting as antagonist) and M_2 was the measured net moment generated during eccentric hamstring contraction (hamstring contracting as agonist, quadriceps contracting as antagonist).

M_1, M_2, EMG_Q and EMG_H were measured and the pair of equations solved separately for any 0.01° knee joint angle in the entire ROM to yield the EMG-to-moment constants K_1 and K_2 separately at each 0.01° knee joint angle. Based on these EMG-to-moment constants (i.e. two arrays of constants with a time resolution of 1 ms) it was possible to calculate the agonist quadriceps moment ($K_1 \cdot EMG_{Q,agonist}$) and antagonist hamstring moment ($K_2 \cdot EMG_{H,antagonist}$) generated throughout the entire ROM during knee extension involving maximal agonist quadriceps contractions [10].

The VL, VM and RF muscles were each assumed to contribute one-third of the total quadriceps knee extensor moment and the BFcl and the ST were each assumed to contribute one-half of the total hamstring flexor moment. Prior to solving the pair of equations the EMG signals of each muscle were normalized relative to the average amplitude of the respective EMG signal obtained while the muscles contracted as agonists.

Thus, the magnitude of agonist quadriceps muscle moment and antagonist hamstring muscle moments could both be predicted, respectively, at each 0.01° knee angle by solving the above set of equations. Only the antagonist hamstring muscle moments generated during agonist quadriceps contraction were selected for further statistical analysis in the present study.

The calculated antagonist hamstring moments were further expressed as a fraction of the measured net knee extension moments and will be referred to as *antagonist moment fraction*.

2.6. Statistical analysis

For each subject the measured knee extension moment recorded during the trials of isokinetic agonist contraction of the quadriceps (referred to as net extension moment), the calculated antagonist hamstring moment and the normalized antagonist hamstring EMG signals moment were averaged in successive 10° intervals. Ensemble averages for the three samples ACLD, UNI and CON were then calculated using the individual subject means. The ACLD and the UNI samples in the patient group were compared using a *t*-test for paired samples. The ACLD and UNI samples were compared to the CON samples using a *t*-test for unpaired samples. The level of significance was set to 5%.

3. Results

When averaged within subject groups the net knee extension moment was greater in CON compared to ACLD in the proximal ROM from 40° to 90° (Fig. 2A, Table 1). Further, net knee extension moment was significantly greater in CON than in UNI at 80–90° (Table 1).

Antagonist hamstring moment fraction is shown in Fig. 3. In the distal ROM (20° to 50° knee joint angle) antagonist moment fraction was on average 55% greater in ACLD than CON ($p<0.05$) (Fig. 3, Table 2). No differences were observed between ACLD and UNI or UNI and CON, respectively.

The magnitude of antagonist hamstring EMG was on average 65.5% greater in ACLD than CON ($p<0.05$) except at the initial and final ROM of 10–20° and 80–90° (Fig. 4, Table 3). Antagonist hamstring coactivation ranged from 29 to 45% (ACLD), 22–43% (UNI) and 19–29% (CON) (Table 3). In both ACLD subjects and control subjects antagonist coactivation was always greater in the lateral hamstring muscles (BFcl) than in the medial hamstring muscle (ST). The average BFcl EMG was 68% (ACLD), 53% (UNI) and 30% (CON) while the average ST EMG was 8% (ACLD), 11% (UNI) and 10% (CON).

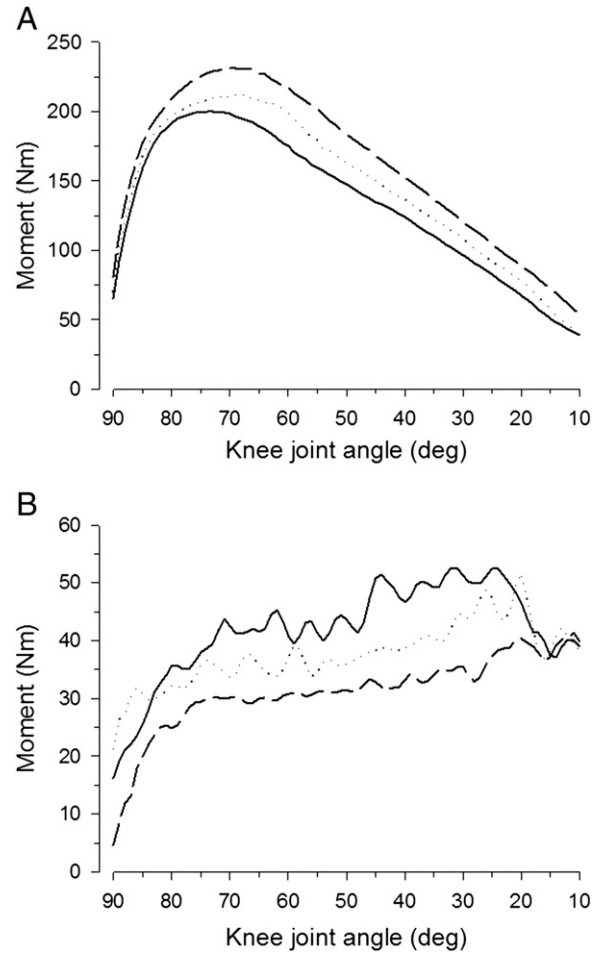

Fig. 2. Measured net extension moments and calculated antagonist moments developed during a concentric quadriceps knee extension (quadriceps agonist, hamstring antagonist). A: Averaged recorded net knee extension moment (Nm) for ACLD (n = 7, solid line), UNI (n = 7, dotted line) and CON (n = 16, dashed line) samples. B: Calculated antagonist hamstring moments (Nm) averaged for ACLD (n = 7, solid line), UNI (n = 7, dotted line) and CON (n = 16, dashed line) samples. For differences between groups, see text (Table 1, Results).

4. Discussion

The main finding of the present study was that antagonist coactivation of the hamstring muscles was significantly increased in the ACL deficient legs compared to a healthy control group during maximal isolated dynamic knee extension. Secondly, in the proximal ROM (40–90°) the recorded net knee extension moment was larger in healthy control legs than ACLD legs. Finally, the antagonist hamstring moment fraction of the measured net knee extension moment increased steeply as the knee approached full extension both in healthy and ACL deficient subjects, however ACL deficient subjects demonstrated a significantly larger antagonist moment fraction in the terminal range of motion (i.e. at 20–50°) compared to non-injured legs. Collectively, the present data support the initial study hypothesis that antagonist hamstring coactivation would increase towards full knee extension, and that antagonist coactivation would be elevated in ACL deficient subjects.

EMG recordings have previously been used to estimate the magnitude of antagonist muscle coactivation during maximal and submaximal isolated knee extension both in healthy [10,11,31] and in ACL deficient subjects [27,32]. Although antagonist hamstring moments have been estimated using cadaver models [4–6], only a few studies have tried to quantify the magnitude of antagonist moment generated in vivo by means of hamstring coactivation during dynamic knee extension in healthy subjects [10,12,23] and in ACL deficient subjects [22]. In contrast to the present findings, Bryant et al. [22] observed no differences in the hamstring antagonist moment during isokinetic knee extension between ACL deficient, ACL reconstructed and healthy subjects, respectively, except in the proximal ROM (70–80° knee flexion) where lower antagonist hamstring moments were observed in the healthy subjects. This apparent disparity may be due to differences between our EMG–moment model and the model used by Bryant et al. [22] As earlier mentioned it is possible that Bryant et

Table 1
Measured net knee extension moment and estimated antagonist muscle moments obtained during quadriceps agonist contraction (averages of 10° intervals, values are mean ± SEM (confidence interval)) in the ACL deficient leg (ACLD, n = 7), contralateral uninjured leg (UNI, n = 7) and in healthy controls (CON, n = 16).

	Measured knee extension moment (Nm)			Antagonist H moment (Nm)		
Angle	ACLD	UNI	CON	ACLID	UNI	CON
10–20°	64.0 ± 9.2 (41.4–86.5)	73.0 ± 7.1 (55.6–90.5)	72.1 ± 3.6 (64.5–79.7)	44.5 ± 7.7 (25.7–63.3)	43.4 ± 7.5 (25.0–61.8)	39.2 ± 4.5 (29.8–48.8)
20–30°	93.7 ± 7.9 (74.5–113.0)	104.8 ± 7.5 (86.4–123.1)	104.8 ± 4.4 (95.3–114.2)	51.2 ± 7.9 (31.8–70.6)	44.3 ± 8.5 (23.5–65.2)	36.9 ± 4.9 (26.4–47.3)
30–40°	120.9 ± 6.4 (105.3–136.5)	133.4 ± 8.6 (112.4–154.5)	137.0 ± 5.3 (125.5–148.3)	49.1 ± 8.5 (28.4–69.9)	39.4 ± 7.3 (21.4–57.3)	34.2 ± 4.5 (24.5–43.9)
40–50°	145.0 ± 5.4 (131.8–158.1)	160.7 ± 12.1 (131.0–190.5)	168.2 ± 6.2 (155.1–181.3)[a]	44.5 ± 9.5 (21.2–67.7)	37.0 ± 5.7 (19.0–54.9)	32.1 ± 4.0 (23.5–40.7)
50–60°	171.7 ± 6.1 (156.7–186.7)	193.0 ± 18.2 (148.4–237.6)	200.6 ± 7.1 (185.4–215.7)[a]	42.2 ± 9.7 (18.5–65.9)	35.6 ± 5.1 (23.3–48.0)	31.0 ± 3.8 (22.8–39.1)
60–70°	195.5 ± 4.1 (185.5–205.6)	209.9 ± 19.9 (161.1–258.6	226.5 ± 7.8 (209.4–242.7)[a]	41.5 ± 9.1 (19.3–63.7)	35.7 ± 5.8 (21.6–49.8)	29.9 ± 3.8 (21.7–38.1)
70–80°	190.7 ± 7.3 (172.8–208.6)	196.8 ± 15.9 (158.0–235.6)	223.3 ± 8.7 (204.8–241.7)[a]	34.6 ± 7.1 (17.2–51.9)	32.3 ± 5.2 (19.7–45.0)	28.6 ± 3.7 (20.7–36.4)
80–90°	125.0 ± 10.8 (98.6–151.3)	133.3 ± 6.5 (117.3–149.3)	164.9 ± 8.6 (146.6–183.2)[a,b]	22.2 ± 3.3 (14.1–30.3)	28.3 ± 4.8 (16.6–39.9)	17.9 ± 2.8 (15.8–27.9)

[a] Statistically significant difference between ACLD and CON ($p<0.05$).
[b] Statistically significant difference between UNI and CON ($p<0.05$).

al. [22] may have underestimated the hamstring antagonist moment as their EMG–moment model was based on EMG calibrations measured during concentric agonist hamstring contraction and not eccentric hamstring contractions as performed in the present study. In support of this notion, hamstring antagonist moment values expressed relative to the recorded net knee extension moment ranged from ~5 to 20% in the study of Bryant et al. [22], while much higher fractions were observed (~13–73%) in the present study. In addition, our results showed that antagonist hamstring moment markedly increased towards full knee extension in contrast to the descending–ascending antagonist hamstring moment curve pattern reported by Bryant et al. [22] The antagonist moment–angle curve produced by the hamstrings in the present study (Fig. 2) strongly indicates that the hamstring muscles serve as stabilizers (i.e. creating posterior tibial drawer) when approaching terminal knee extension, at which ROM the quadriceps muscle also demonstrates the greatest potential to cause anterior translation and internal rotation of the tibia [5,33]. Excessive translation and rotation of the tibia may result in subluxation of the knee joint and/or injuries of other knee joint structures in the ACL deficient knee [8]. The elevated antagonist hamstring moment fraction observed in the ACL deficient subjects in the present study may therefore likely reflect a compensatory neuromuscular adaptation to counteract the increased laxity of the knee joint. This notion is supported by modeling data by Yangawa et al. [7], which confirms that coactivation of the hamstring muscles during isolated dynamic (isokinetic) knee extension effectively reduces ATT.

It is important to notice that antagonist coactivation was observed both in the ACL deficient subjects as well as in healthy control subjects in the present study. However, the magnitude of antagonist hamstring coactivation and antagonist moment fraction was more than two fold larger in the ACL deficient subjects. It may be speculated that the control of antagonist coactivation is not only regulated by sensory input from mechanoreceptors in the knee joint capsule but likely also is modulated by central motor programming. The finding that antagonist coactivation was present in both the control subjects and ACL deficient subjects supports previous studies that show that the coordination of antagonistic muscles, in general, is centrally controlled [34–36]. Animal and human studies have shown that the magnitude of neuromuscular activity in the hamstring muscles is modulated by neural input from the cruciate ligaments [9,12,37,38] and that this modulation is absent following surgical ACL reconstruction [39]. However, the elevated levels of antagonist hamstring coactivation in the ACL deficient legs observed in the present study indicate that the amount of antagonist coactivation is not solely controlled by knee ligament reflex arcs, which concurs with previous findings [22,40].

The elapsed time from the ACL injury to the day of testing was on average 47.5 ± 54.2 (range 6–144) months. It is therefore possible that the ACL subjects had adapted to their injury by reprogramming the coactivation pattern of the knee flexor muscles, in order to increase stability of the knee joint. Future investigations of antagonist coactivation in subjects with acute ACL rupture and following short-term recovery (6–12 months) are necessary to confirm whether the observed increase in antagonist muscle coactivation represents a long-term adaptation.

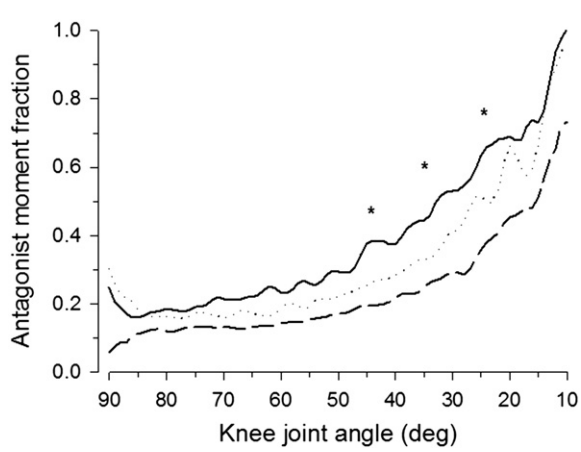

Fig. 3. Antagonist hamstring moment expressed as a fraction of the measured net extension moment. Averaged for ACLD (n = 7), UNI (n = 7) and CON (n = 16) samples.- Solid, dotted and dashed lines represent ACLD, UNI and CON, respectively.* Indicates statistically significant difference between the ACLD and CON samples.

Table 2
Antagonist hamstring moment fraction of measured net extension moment (averages of 10° intervals, values are mean ± SEM (confidence interval) in ACL deficient legs (ACLD, n = 7), contralateral uninjured legs (UNI, n = 7) and in healthy controls (CON, n = 16).

Antagonist moment fraction			
Angle	ACLD	UNI	CON
10–20°	0.73 ± 0.10 (0.49–0.98)	0.62 ± 0.12 (0.33–0.92)	0.56 ± 0.07 (0.42–0.70)
20–30°	0.56 ± 0.08 (0.35–0.76)	0.44 ± 0.08 (0.23–0.64)	0.36 ± 0.05 (0.25–0.46)[a]
30–40°	0.40 ± 0.07 (0.24–0.56)	0.30 ± 0.06 (0.16–0.45)	0.26 ± 0.03 (0.18–0.33)[a]
40–50°	0.30 ± 0.06 (0.15–0.45)	0.24 ± 0.04 (0.13–0.35)	0.19 ± 0.02 (0.14–0.25)[a]
50–60°	0.24 ± 0.06 (0.11–0.38)	0.19 ± 0.03 (0.11–0.28)	0.16 ± 0.02 (0.11–0.20)
60–70°	0.21 ± 0.04 (0.10–0.32)	0.18 ± 0.04 (0.09–0.27)	0.14 ± 0.02 (0.10–0.18)
70–80°	0.18 ± 0.04 (0.09–0.27)	0.17 ± 0.03 (0.10–0.24)	0.13 ± 0.02 (0.09–0.17)
80–90°	0.19 ± 0.03 (0.12–0.25)	0.21 ± 0.03 (0.13–0.29)	0.14 ± 0.02 (0.09–0.18)

[a] Statistically significant difference between ACLD and CON ($p<0.05$).

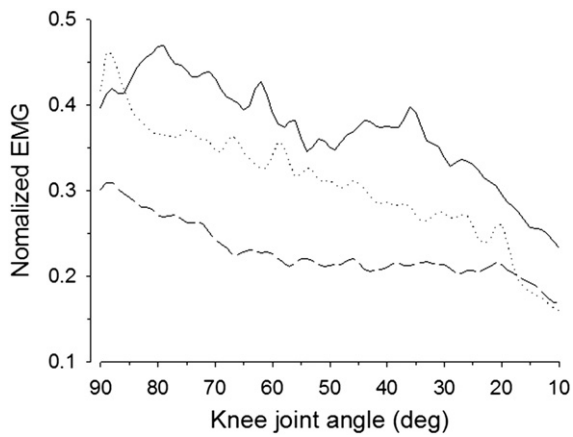

Fig. 4. Normalized antagonist hamstring EMG (BFcl, ST) recorded during knee extension. Averaged for ACLD (n = 7), UNI (n = 7) and CON (n = 16) samples. Solid, dotted and dashed lines represent ACLD, UNI and CON, respectively.

The antagonist EMG was dominated by the lateral hamstring muscle (BFcl). Because the internal/external rotation axis passes through the medial part of the knee joint the BFcl muscle becomes a powerful external rotator because of its large moment arm [41], which increases with as the knee flexes [42]. Therefore, the differential hamstring response seems to serve to prevent excessive tibial internal rotation during knee extension in addition to creating a knee flexion moment that opposes anterior tibial drawer.

The measured net extension moment was larger in healthy legs than in the ACLD legs in the 40–90° ROM and in UNI legs at 80–90° ROM. These findings may be explained by the high generation of antagonist hamstring moment in ACL deficient legs. In strong support of this notion, the calculation of the agonist quadriceps moment (i.e. adding the magnitude of antagonist hamstring moment to the measured net knee extension moment) revealed no statistically significant differences between the control subjects and the ACL deficient subjects (data not shown).

Gait analyses have shown that ACL subjects develop lower net knee extensor moments during the stance phase compared to healthy subjects, and the phenomenon has been referred to as *quadriceps avoidance gait* [43,44]. However, increased coactivation of the hamstring muscles in ACL deficient subjects could also reduce the net knee extensor moment during gait. This certainly questions the existence of a quadriceps avoidance pattern and encourages further investigation of antagonist coactivation in ACL subjects in more natural types of movement.

Several studies have reported that ACL deficient copers and non-copers may demonstrate different movement patterns [25,26,45,46]. Adequate hamstring coactivation [25,26,46] and a retained quadriceps muscle function [47] have been suggested as critical factors in the differential response to ACL injury. The ACL deficient subjects included in this study could not be classified as either pure copers or non-copers [45,48] but according to their Tegner and Lysholm scores the knee function ranged from “poor” to “excellent” [28,29]. Despite the wide individual range, the magnitude of hamstring coactivation observed in the present ACL deficient subjects was not related to their self-rated knee score. Thus, increased hamstring coactivation during maximal knee extension may represent a general response to ACL deficiency. However, further investigations of hamstring coactivation and quadriceps muscle function in ACL deficient copers and non-copers are needed to confirm this notion.

5. Conclusions

Our initial hypothesis that hamstring coactivation would increase towards terminal knee extension, and that coactivation would be elevated in ACL deficient individuals was confirmed by the present results. No significant differences were observed in terms of antagonist coactivation between the injured and uninjured leg of the ACL deficient subjects or between the uninjured leg and the leg of the control subjects. However, a larger antagonist moment fraction and greater antagonist EMG activity were observed in ACL deficient legs in distal ROM when compared to healthy controls. The results suggest that antagonist coactivation is an integrated element of the central motor program for knee extension. The observed coactivation pattern of the knee flexors during maximal knee extension movements indicates that the hamstring muscles play an important role in stabilizing the knee joint and that this effect seems to be augmented in ACL deficient subjects, which may represent an adaptation mechanism to prevent excessive ATT. Future investigations with the present EMG–moment model may enhance our understanding of the specific adaptations involved with acute and long-term ACL deficiency. Finally, the present study provides a method to investigate e.g. effects of treatment/rehabilitation on the hamstring coactivation during dynamic knee extension in ACL deficient subjects.

Table 3
Normalized antagonist hamstrings EMG (averages of 10° intervals, values are mean ± SEM (confidence interval)) in ACL deficient legs (ACLD, n = 7), contralateral uninjured legs (UNI, n = 7) and in healthy controls (CON, n = 16).

Antagonist EMG			
Angle	ACLD	UNI	CON
10–20°	0.29 ± 0.06 (0.14–0.44)	0.22 ± 0.04 (0.12–0.32)	0.19 ± 0.02 (0.14–0.24)
20–30°	0.34 ± 0.07 (0.18–0.51)	0.27 ± 0.05 (0.14–0.39)	0.21 ± 0.03 (0.15–0.27)[a]
30–40°	0.38 ± 0.07 (0.22–0.54)	0.28 ± 0.06 (0.13–0.44)	0.21 ± 0.03 (0.16–0.27)[a]
40–50°	0.36 ± 0.06 (0.20–0.52)	0.31 ± 0.07 (0.14–0.48)	0.21 ± 0.03 (0.15–0.27)[a]
50–60°	0.39 ± 0.07 (0.21–0.57)	0.33 ± 0.06 (0.18–0.48)	0.22 ± 0.03 (0.16–0.27)[a]
60–70°	0.42 ± 0.07 (0.24–0.60)	0.35 ± 0.08 (0.17–0.54)	0.23 ± 0.03 (0.17–0.29)[a]
70–80°	0.45 ± 0.09 (0.24–0.66)	0.37 ± 0.08 (0.18–0.56)	0.26 ± 0.04 (0.19–0.34)[a]
80–90°	0.42 ± 0.07 (0.24–0.59)	0.43 ± 0.10 (0.17–0.68)	0.29 ± 0.04 (0.20–0.38)

[a] Statistically significant difference between ACLD and CON ($p<0.05$).

Conflict of interest

None of the authors have any financial or personal relationships that would be deemed a conflict of interest.

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